Umur Yıldız

+90 530 182 2656 | umuryildiz_@outlook.com | linkedin.com/in/umuryildiz | github.com/UYildiz12

Education

Bilkent University — Ankara, Turkey
Master of Science in Neuroscience — *Sep. 2025 – Sep. 2027 (Expected)*

Bilkent University — Ankara, Turkey
Bachelor of Arts in Psychology, Minor in Software Development — *Jan. 2020 – Jan. 2024*

Neuromatch Academy Summer School — Remote
Computational Neuroscience (2023), Deep Learning (2024), NeuroAI (2025) — *Summer 2023, 2024, 2025*

Experience

Machine Learning Engineer
Cybersound Teknoloji — *Present*
- Develop machine learning systems for audio applications within a TÜBİTAK 1507 R&D project, covering data collection, cleaning and annotation, model development and training, evaluation, and integration.

Teaching Assistant & Lab Tutor — Ankara, Turkey
Bilkent University — *Aug. 2023 – Present*
- Began as a Lab Tutor and Recitation Instructor for CS 115 and CS 125 (undergraduate programming and data-analysis courses), supporting more than 110 students.
- Responsibilities expanded during M.Sc. studies to provide teaching, laboratory, and recitation support across neuroscience and psychology courses, including research methods, programming, and data analysis.

Contract AI Data Evaluator & Software Developer — Remote
DataAnnotation, Outlier AI, Appen, and other platforms — *May 2022 – July 2024; Sep. 2024 – Present*
- Evaluated AI outputs involving scientific frameworks, assessing technical accuracy and the quality of scientific reasoning.
- Develop long-horizon software development and machine learning tasks for benchmarking AI coding agents, each with a containerized environment, reference solution, and automated verifier, calibrated for difficulty through agent trials and hardened against cheating.

Publications

Yıldız, U., & Ürgen, B. A. (2026). *Naturalistic stimulus reconstruction from fMRI: A primer in the Natural Scenes Dataset. Aperture Neuro, 6.* doi.org/10.52294/001c.162412

Yıldız, U., Yüce, B., Karadağ, A., Pekçetin, T. N., & Ürgen, B. A. (2026). *Show Your Work: Mechanistic Transparency in LLM-Powered Human-Robot Interaction. HRI '26 Companion.* doi.org/10.1145/3776734.3794439

Yıldız, U., & Ürgen, B. A. (2025). *Multiarrangement: A Plug & Play Geometric Data Collection Package for Video Stimuli* [Preprint; under review]. doi.org/10.1101/2025.10.17.683124

Awards & Honors

TÜBİTAK 2247-C STAR Young Researcher Scholarship
Full Scholarship for B.A. and M.Sc. studies at Bilkent University
High Honors Graduate (Psychology & Software Development)

Technical Skills

Languages: Turkish (native); English (IELTS 8.0; CEFR C1)
Programming Languages: Python, C++, MATLAB, SQL, TypeScript
Data Science & ML: PyTorch, Keras, Pandas, NumPy, Scikit-learn, Hugging Face Diffusers, CLIP, SDXL, DSPy
Neuroscience Tools: PyMVPA, CoSMoMVPA, FSL, SPM
Data Visualization: Matplotlib, Seaborn, Plotly, Tableau